

Tony Nguyen

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Education

M.S. in Artificial Intelligence, Machine Learning
Northeastern University

Expected May 2028
Seattle, WA

B.A. in Computer Science, Data Science & Economics
B.B.A. in Business Administration, Finance

May 2026
Spokane, WA

Gonzaga University

President's Honor List, *Summa Cum Laude*, GPA: 3.96

Research Interests

Knowledge representation and evaluation in LLMs. Skill integration for AI-assisted threat hunting. Retrieval-augmented generation, agentic systems, and resource-constrained deployment for reliable AI.

Research & Industry Experience

AI / Machine Learning Research Assistant (Advisor: Dr. Jay Yang)
Institute for Informatics & Applied Technology - Gonzaga University

January 2025 - May 2026
Spokane, WA

- Designed and built the **Wazuh MCP Server**, an open-source Python server (**30+ endpoints**) enabling LLMs to query, investigate, and analyze security events from a Wazuh SIEM platform through structured tool calls.
- Designed a controlled **benchmarking study** comparing **skill integration** across two frontier models (Claude Sonnet and Opus) against a **ground-truth attack chain**, showing that structured domain knowledge closes the performance gap between smaller and larger models, and extended the study to **Gemma 4 (31B)** on local GPU infrastructure to test generalization to open-source models.
- Benchmarked **three LLMs** on **more than 60 CVE reports** using a **three-stage RAG pipeline** with specialized **agents** for retrieval, analysis, and remediation recommendation, focusing on insightfulness and verifiability.
- Developed and delivered a workshop on **RAG** systems and **LLM evaluation** to over **30 participants**, covering classical metrics, agent-based evaluation, and responsible AI practices.

Data Engineer Intern
KPIM Vietnam

June 2025 - September 2025
Hanoi, Vietnam

- Designed and maintained **10+ data pipelines** using **SQL, Python, and Microsoft Fabric** for **2 Big Data projects**, enforcing complex schema validation, data quality checks, and optimizations to make querying **50% faster** than our client's internal system.
- Built reusable tooling and documentation to streamline troubleshooting, standardize workflows, and support stakeholders with reliable, self-serve financial and operational data of the client's **6 hotel properties**.

Data Engineer - Innovation Intern
KONE Elevators & Escalators

June 2024 - August 2024
Singapore

- Built a **centralized Customer 360** by integrating **20 years** of data from **5 countries**; aligned models to enterprise architecture and recommended schema optimizations to improve consistency and join performance.
- Developed a **Tableau** dashboard that fed from **SAP, Salesforce, and Excel**; created data models and calculations to monitor customer behavior, service KPIs, and revenue in real time.
- Engineered and operated **ETL** pipelines on **AWS S3** for ingestion, storage, and validation; automated integrity checks and partitioning to **reduce data retrieval latency by 40%** and improve reliability.

Data Science & Machine Learning Intern
TecAlliance Vietnam

June 2023 - August 2023
Ho Chi Minh City, Vietnam

- Engineered Python algorithms to parse data from **100+ Wiring Diagram SVGs**, creating structured datasets for computer vision models (**YOLO**) and advanced ML pipelines.
- Deployed a **Neo4j graph database** to store complex relationships extracted from automotive schematics, increasing **query speed by 80%** compared to the client's current database and providing deeper insights.
- Led weekly **global** team meetings of nearly **15 people** to unify data engineering efforts, consolidate findings, and manage development timelines across multiple time zones.

Conference Poster Presentations

- Value & Responsibility in AI Technologies Conference, 2026. Spokane, WA.
- University of Washington Undergraduate Research Symposium, 2026. Seattle, WA.
- Value & Responsibility in AI Technologies Conference, 2025. Spokane, WA.
- Spokane Intercollegiate Research Conference, 2025. Spokane, WA.

Projects

- Minecraft AI Companion – Senior Capstone Project

August 2025 - May 2026

 - Built an open-source Minecraft mod using a **two-model architecture** (local LLM for real-time dialogue, cloud model for retrieval-augmented queries), designing the context pipeline, tool-calling interface, and content guardrails for safe in-game interactions.
- Mortality Risk Simulator

January 2025 - May 2025

 - Led development of a **neural-network-based mortality risk prediction pipeline** over demographic, vitals, and medical history features to support more accurate and fair underwriting review.

Teaching & Mentoring Experience

- Teaching Assistant

August 2024 - May 2026

Department of Computer Science - Gonzaga University

Spokane, WA

 - Supported **150+** undergraduates across **3 data science courses**, leading weekly office hours to reinforce students' Python, pandas, and scikit-learn skills and core algorithms and data workflows.
 - Graded **10 programming assignments each term** using shared rubrics, providing targeted feedback on correctness, code structure, and documentation to improve code quality and conceptual understanding.
- Academic Tutor & Mentor

October 2022 - May 2026

Gonzaga Global - Gonzaga University

Spokane, WA

 - Provided tutoring and lab support for **7 computer science and quantitative courses** for **dozens of students each term**, contributing to **15%+ average grade improvement** for tutees.
 - Mentored international students on adapting to U.S. college coursework and expectations, including study strategies, communication with professors, and academic integrity.

Professional & Leadership Experience

- Administrative Assistant

March 2022 - May 2026

Center for Global Engagement - Gonzaga University

Spokane, WA

 - Mentored **50+** incoming international students through cultural and academic transition to US college; designed automated workflows reducing administrative burden by **30%**, enabling more direct student support.
- Co-Founder & Senior Advisor

November 2021 - May 2025

Vietnamese Student Association - Gonzaga University

Spokane, WA

 - **Co-founded** the organization to foster supportive community for Vietnamese students; **mentored 10+ leaders** in data-informed planning, increasing event turnout by **20%** and generating **\$700+ profit**.

Professional Affiliations & Memberships

- **Member**, *Society of Hispanic Professional Engineers*

Technical Proficiencies

- Programming Languages

Python, SQL (MySQL, BigQuery SQL), C++, Java.
- Data & ML Frameworks

pandas, NumPy, scikit-learn, PyTorch, spaCy, Streamlit, Flask, FastAPI, pytest.
- Data Engineering & Analytics

Tableau, Power BI, Microsoft Fabric, Salesforce CRM, ETL pipelines, Docker, Git, Linux, Google Cloud, AWS.
- AI & LLM Systems

RAG pipelines, LangChain, CrewAI, Semantic Kernel, Hugging Face Transformers, MCP, AI agents, embeddings.